

Appendix B: Causal discovery as a method to determine Average Causal Effect of Payment Protection Plans on Default

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1 Appendix B: Markov Blanket and DAGs

1.1 ($\alpha = 5\%$)

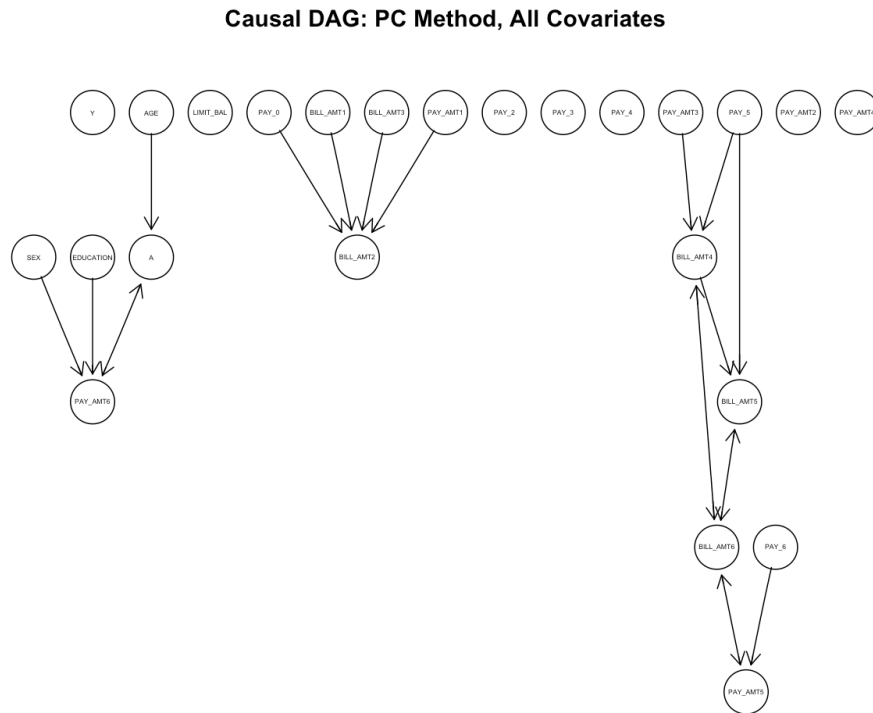


Figure 1: Full DAG for the PC with all covariates model.

Consensus DAG (Edges in >50% of Bootstraps): PC Method, All Covariates

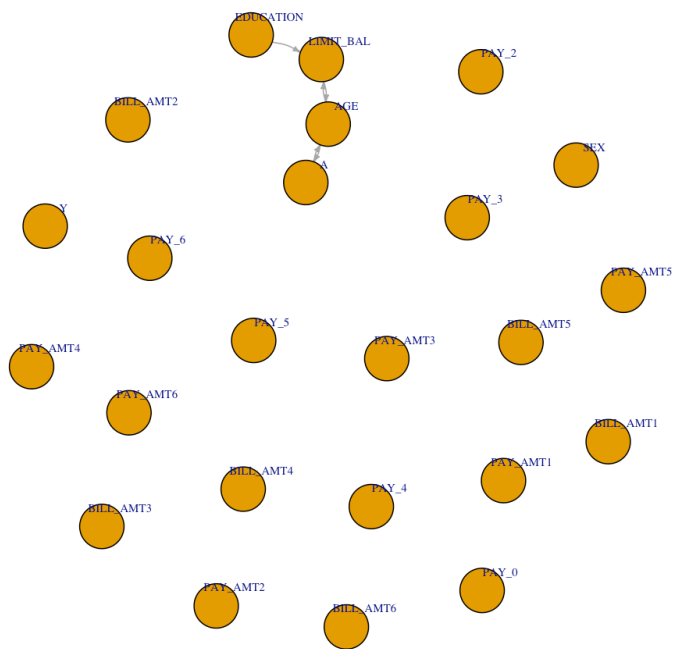


Figure 2: Consensus DAG (edges included if they are in over 50% of the bootstraps) for the PC with all covariates model.

Causal DAG: PC Method, Limited Covariates

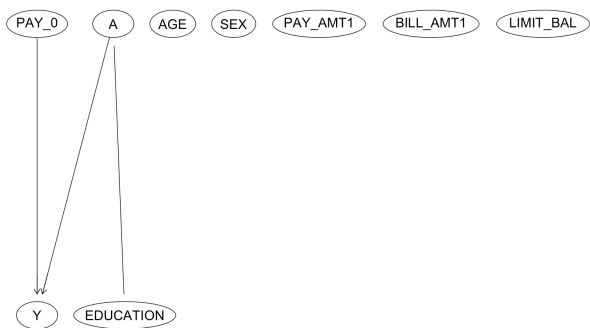


Figure 3: Full DAG for the PC with limited covariates model.

Causal DAG: Rank PC Method

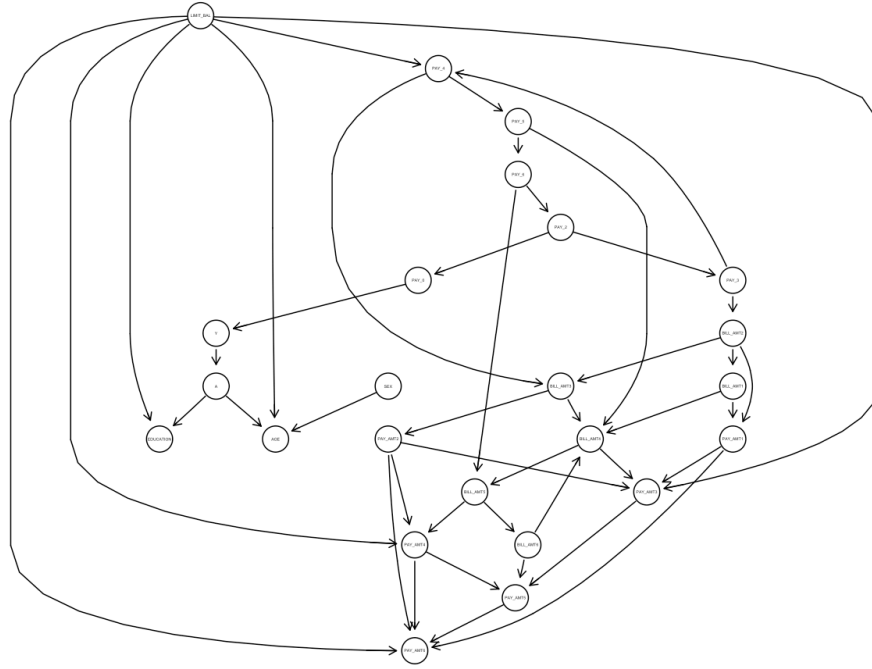


Figure 4: Full DAG for the Rank PC model.

boot_blankets_local_unlisted							
A	PAY_0	PAY_3	PAY_2	BILL_AMT1	EDUCATION	PAY_AMT4	PAY_AMT2
0.993	0.992	0.628	0.374	0.281	0.270	0.249	0.243
PAY_AMT3	AGE	PAY_AMT1	LIMIT_BAL	BILL_AMT6	PAY_4	SEX	PAY_AMT5
0.216	0.163	0.135	0.110	0.089	0.068	0.062	0.057
BILL_AMT2	PAY_5	BILL_AMT3	PAY_6	BILL_AMT5	PAY_AMT6	BILL_AMT4	
0.056	0.050	0.044	0.042	0.039	0.024	0.018	

Figure 5: Local Discovery (Inter-IAMB, Ranked Data): Bootstrapped Markov Blanket for Y

boot_blankets_local_unlisted_a							
AGE	Y	EDUCATION	SEX	PAY_0	LIMIT_BAL	PAY_AMT4	BILL_AMT2
1.000	0.990	0.857	0.425	0.304	0.264	0.201	0.170
PAY_AMT3	BILL_AMT1	PAY_AMT6	PAY_AMT2	BILL_AMT6	BILL_AMT5	PAY_AMT1	PAY_AMT5
0.139	0.131	0.128	0.109	0.082	0.068	0.068	0.066
PAY_3	PAY_4	PAY_2	PAY_6	BILL_AMT3	PAY_5	BILL_AMT4	
0.062	0.060	0.058	0.046	0.043	0.042	0.017	

Figure 6: Local Discovery (Inter-IAMB, Ranked Data): Bootstrapped Markov Blanket for A

boot_blankets_rank_pc_unlisted							
PAY_0	A	PAY_3	PAY_2	PAY_5	PAY_AMT2	AGE	PAY_6
0.980	0.978	0.499	0.344	0.192	0.177	0.157	0.146
PAY_AMT4	PAY_AMT3	PAY_4	PAY_AMT1	EDUCATION	LIMIT_BAL	SEX	PAY_AMT5
0.063	0.053	0.041	0.036	0.029	0.020	0.014	0.001

Figure 7: Rank PC: Bootstrapped Markov Blanket for Y

boot_blankets_rank_pc_unlisted_a							
Y	AGE	EDUCATION	PAY_3	LIMIT_BAL	PAY_0	PAY_AMT2	PAY_AMT4
0.978	0.954	0.626	0.296	0.210	0.183	0.140	0.103
SEX	PAY_AMT3	PAY_AMT6	PAY_AMT1	PAY_AMT5	PAY_2	PAY_4	BILL_AMT1
0.072	0.053	0.033	0.017	0.006	0.003	0.003	0.001
BILL_AMT6	PAY_5						
0.001	0.001						

Figure 8: Rank PC: Bootstrapped Markov Blanket for A

boot_blankets_pc_unlisted			
PAY_0	A	EDUCATION	SEX
0.993	0.945	0.204	0.064

Figure 9: PC Limited Covariates: Bootstrapped Markov Blanket for Y

boot_blankets_pc_a_unlisted					
Y	PAY_0	EDUCATION	SEX	AGE	LIMIT_BAL
0.945	0.719	0.448	0.075	0.002	0.001

Figure 10: PC Limited Covariates: Bootstrapped Markov Blanket for A

boot_blankets_pc_unlisted_full							
LIMIT_BAL	AGE	BILL_AMT6	PAY_AMT1	PAY_AMT6	PAY_AMT4	PAY_3	BILL_AMT3
0.075	0.072	0.040	0.015	0.015	0.013	0.009	0.008

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BILL_AMT4	PAY_AMT2	A	PAY_AMT5	PAY_AMT3	BILL_AMT1	BILL_AMT2	BILL_AMT5
0.008	0.007	0.006	0.006	0.005	0.004	0.004	0.002
PAY_5	EDUCATION	PAY_0					
0.002	0.001	0.001					

Figure 11: PC All Covariates: Bootstrapped Markov Blanket for Y

boot_blankets_pc_a_unlisted_full								
AGE	PAY_AMT6	LIMIT_BAL	EDUCATION	PAY_AMT1	Y	PAY_0	PAY_AMT3	
0.506	0.038	0.031	0.022	0.006	0.006	0.004	0.003	
PAY_AMT2	BILL_AMT1	BILL_AMT2	PAY_4	PAY_AMT4	PAY_AMT5			
0.002	0.001	0.001	0.001	0.001	0.001			

Figure 12: PC All Covariates: Bootstrapped Markov Blanket for A

1.2 ($\alpha = 1\%$)

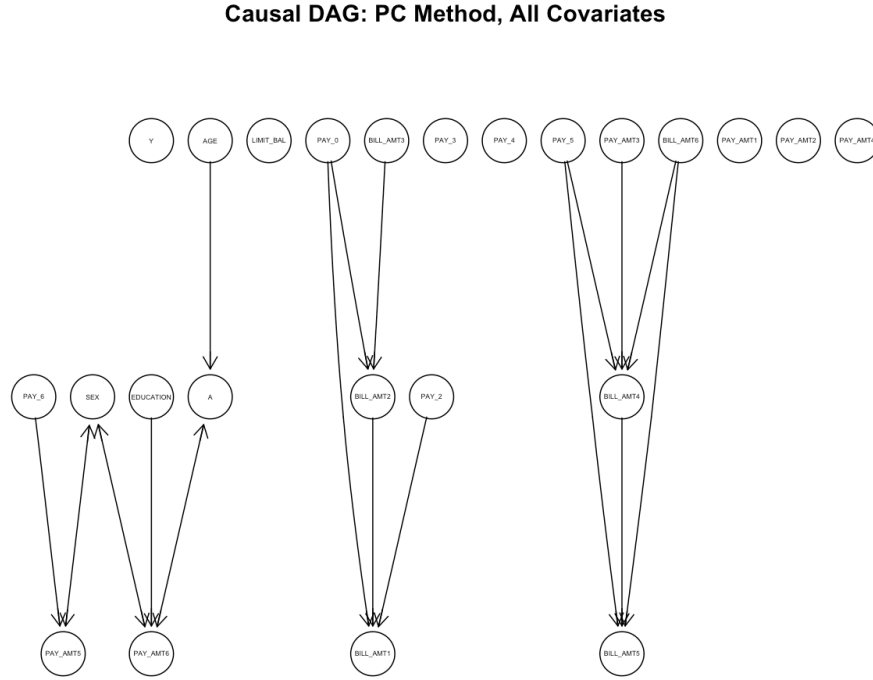
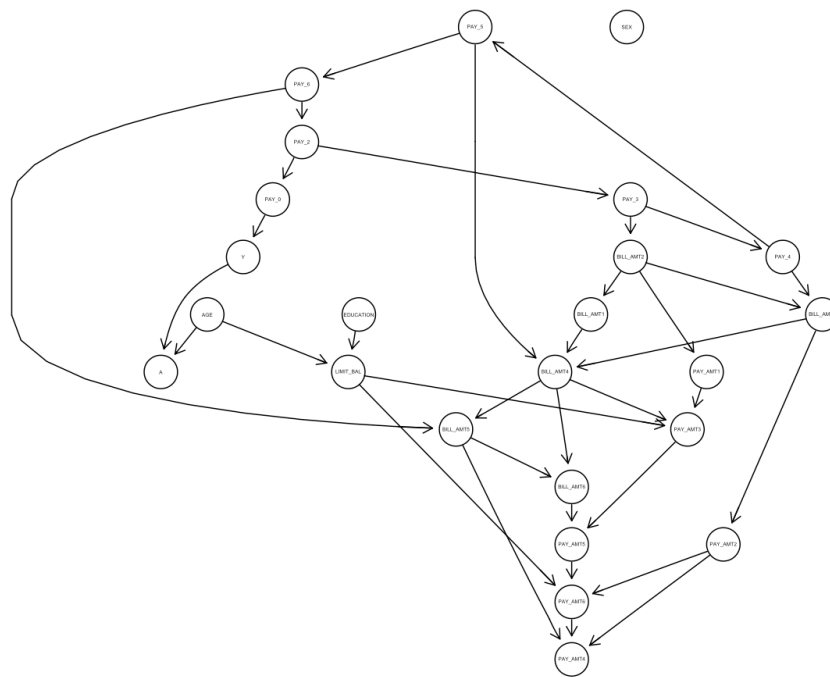


Figure 13: Full DAG for the PC with all covariates model.

Causal DAG: Rank PC Method



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boot_blankets_local_unlisted							
A	PAY_0	PAY_3	PAY_AMT2	PAY_AMT4	BILL_AMT1	PAY_AMT3	PAY_2
0.987	0.950	0.377	0.177	0.177	0.158	0.141	0.114
EDUCATION	LIMIT_BAL	PAY_AMT1	AGE	PAY_4	PAY_5	PAY_AMT5	BILL_AMT6
0.107	0.061	0.057	0.051	0.029	0.021	0.020	0.019
BILL_AMT5	BILL_AMT2	SEX	PAY_6	BILL_AMT3	BILL_AMT4	PAY_AMT6	
0.018	0.013	0.013	0.012	0.011	0.008	0.007	

Figure 16: Local Discovery (Inter-IAMB, Ranked Data): Bootstrapped Markov Blanket for Y

boot_blankets_local_unlisted_a							
AGE	Y	EDUCATION	SEX	PAY_0	LIMIT_BAL	PAY_AMT4	PAY_AMT6
1.000	0.936	0.635	0.238	0.180	0.122	0.122	0.073
PAY_AMT3	PAY_4	BILL_AMT2	PAY_2	BILL_AMT1	BILL_AMT5	PAY_AMT2	BILL_AMT6
0.056	0.036	0.030	0.029	0.025	0.025	0.025	0.022
PAY_3	PAY_AMT5	PAY_5	BILL_AMT3	PAY_6	BILL_AMT4	PAY_AMT1	
0.021	0.017	0.016	0.008	0.008	0.007	0.007	

Figure 17: Local Discovery (Inter-IAMB, Ranked Data): Bootstrapped Markov Blanket for A

1.3 ($\alpha = 10\%$)

Causal DAG: PC Method, All Covariates

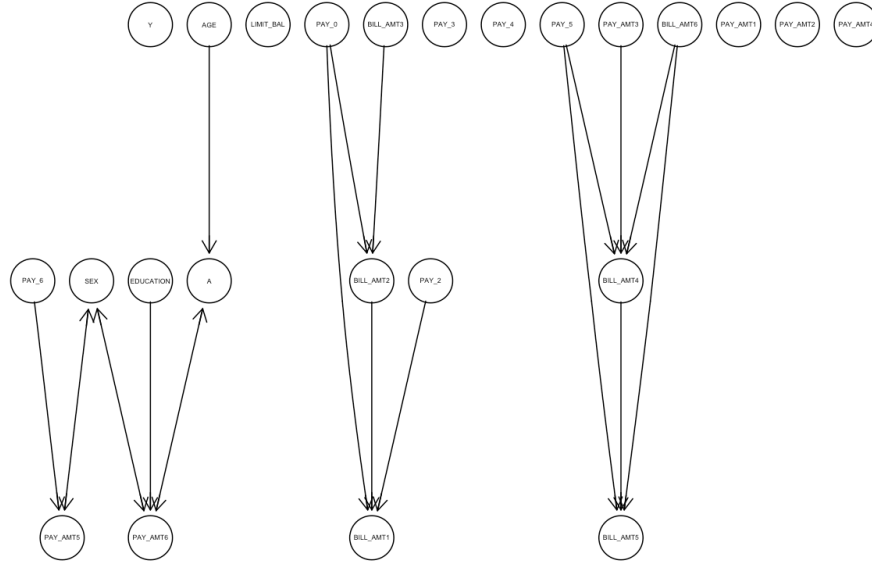
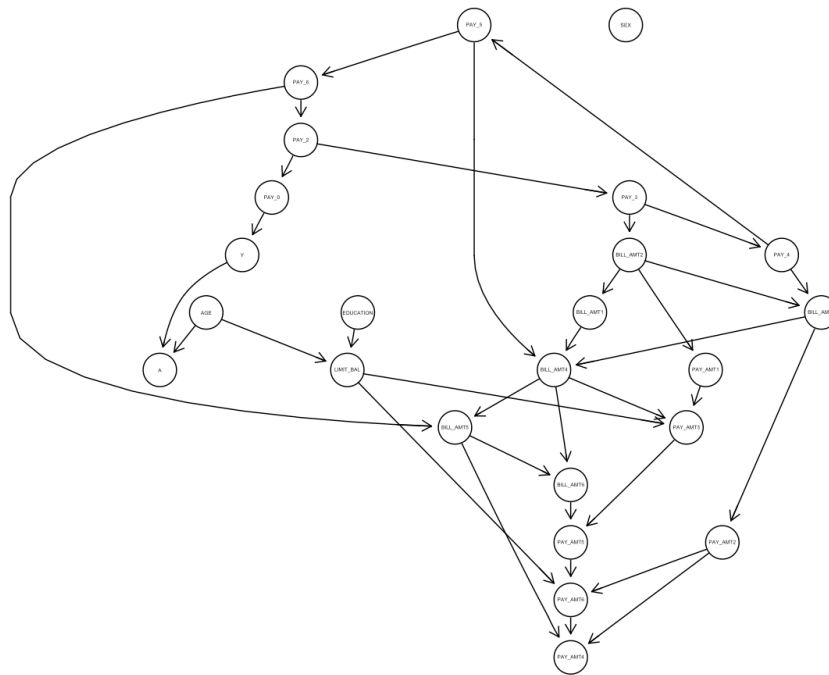


Figure 18: Full DAG for the PC with all covariates model.

Causal DAG: Rank PC Method



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boot_blankets_local_unlisted

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A	PAY_0	PAY_3	PAY_2	EDUCATION	BILL_AMT1	PAY_AMT2	PAY_AMT3
1.000	0.992	0.713	0.494	0.373	0.362	0.287	0.285
PAY_AMT4	AGE	PAY_AMT1	BILL_AMT6	LIMIT_BAL	PAY_AMT5	PAY_6	BILL_AMT2
0.285	0.265	0.232	0.225	0.174	0.113	0.100	0.095
SEX	PAY_5	PAY_4	BILL_AMT3	BILL_AMT5	PAY_AMT6	BILL_AMT4	
0.094	0.084	0.080	0.076	0.075	0.075	0.046	

Figure 21: Local Discovery (Inter-IAMB, Ranked Data): Bootstrapped Markov Blanket for Y

boot_blankets_local_unlisted_a							
AGE	Y	EDUCATION	SEX	PAY_0	LIMIT_BAL	PAY_AMT4	PAY_AMT2
1.000	0.998	0.915	0.535	0.370	0.326	0.245	0.242
BILL_AMT2	BILL_AMT1	PAY_AMT6	PAY_AMT3	BILL_AMT6	BILL_AMT5	PAY_2	PAY_AMT1
0.233	0.222	0.188	0.167	0.137	0.127	0.116	0.110
PAY_3	PAY_4	PAY_AMT5	PAY_6	BILL_AMT3	PAY_5	BILL_AMT4	
0.104	0.102	0.090	0.089	0.076	0.073	0.052	

Figure 22: Local Discovery (Inter-IAMB, Ranked Data): Bootstrapped Markov Blanket for A